

Parallelism and Multiprocessor Systems

1. Introduction

Parallelism is a key concept in **computer architecture** that improves system performance by executing multiple operations simultaneously. Modern computers use parallelism to achieve high performance, efficiency, and scalability.

2. Why Parallelism is Needed

- Higher performance and throughput
 - Better utilization of hardware resources
 - Faster execution of large programs
 - Improved responsiveness
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3. Types of Parallelism

- **Instruction-Level Parallelism (ILP):** Multiple instructions executed simultaneously
 - **Data-Level Parallelism (DLP):** Same operation on multiple data elements
 - **Task-Level Parallelism (TLP):** Different tasks or threads executed in parallel
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4. Flynn's Taxonomy

Category	Description	Example
SISD	Single Instruction, Single Data	Traditional single-core CPU
SIMD	Single Instruction, Multiple Data	GPUs, Vector processors
MISD	Multiple Instruction, Single Data	Rare, fault-tolerant systems
MIMD	Multiple Instruction, Multiple Data	Multiprocessor systems

Diagram: Flynn's Taxonomy

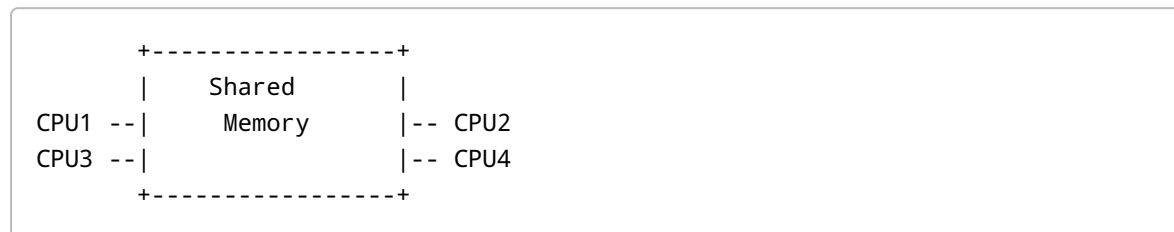
```
Instructions
|
V
SISD ----> Single Data Stream
SIMD ----> Multiple Data Streams
MISD ----> Single Data Stream with Multiple Instructions
MIMD ----> Multiple Data Streams and Instructions
```

5. Parallel Computing Models

5.1 Shared Memory Model

- All processors share a common memory space
- Communication through shared variables

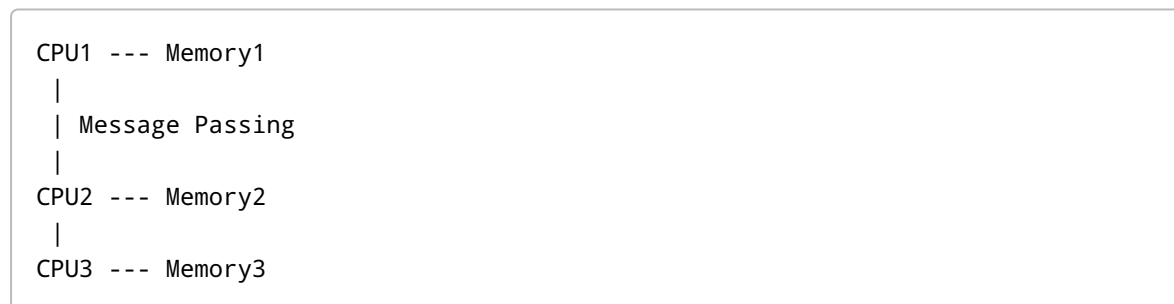
Diagram: Shared Memory Model



5.2 Distributed Memory Model

- Each processor has private memory
- Communication via message passing

Diagram: Distributed Memory Model



6. Multiprocessor Systems

Key Objectives

- Increase performance
- Improve reliability
- Support parallel execution

6.1 Symmetric Multiprocessing (SMP)

- Identical processors sharing memory and OS
- Uniform Memory Access (UMA)

Diagram: SMP System

```
CPU1 ---+
CPU2 ---+--- Shared Memory --- I/O Devices
CPU3 ---+
```

6.2 Non-Uniform Memory Access (NUMA)

- Memory access time depends on memory location
- Each processor has local memory

Diagram: NUMA System

```
CPU1 --- Local Memory1
CPU2 --- Local Memory2
CPU3 --- Local Memory3
      \   |   /
       Shared Interconnect
```

6.3 Asymmetric Multiprocessing (AMP)

- Master processor controls slave processors
- Slaves perform specific tasks

7. Multicore Processors

- Multiple cores on a single chip
- Advantages: higher performance, lower power, efficient parallel execution

8. Interconnection Networks

- Bus-based, Crossbar, Mesh, Ring networks

9. Cache Coherence

- Ensures all processors see consistent data in shared-memory systems
- Protocols: MSI, MESI, MOESI

10. Synchronization in Parallel Systems

- Locks, Semaphores, Barriers
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11. Challenges

- Load balancing, Deadlocks, Race conditions, Communication overhead, Scalability
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12. Advantages

- High performance, Reliability, Scalability

13. Disadvantages

- Complex hardware, Difficult programming, Higher cost
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14. Applications

- Scientific simulations, AI/ML, Databases, Graphics processing
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15. Summary

Parallelism and multiprocessor systems enable concurrent execution, boosting performance and efficiency. Understanding architecture models and memory types is key for computer architecture studies.

End of Lecture Notes with Diagrams